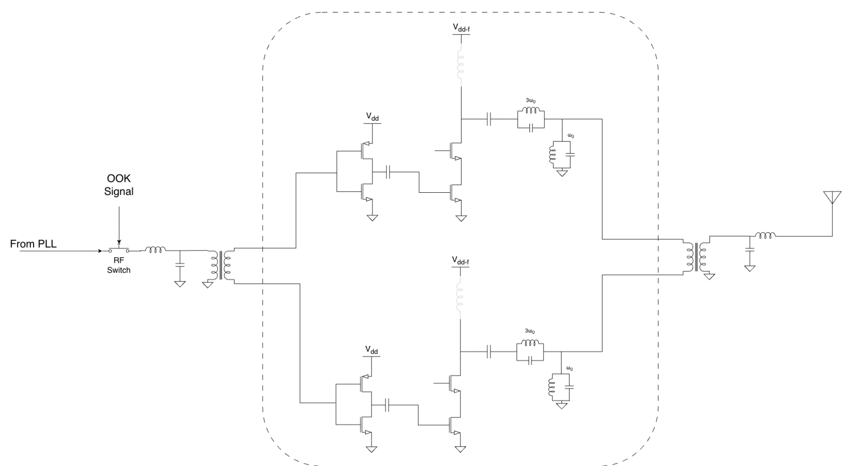
Cornell Custom Silicon Systems

Spring 2026

2.4GHz Class-F Power Amplifier



Technical Design Report

Wali Afridi (wua3) & Taylor Do (txd3)

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1 Abstract

This document details the design, simulation, and tapeout of Cornell Custom Silicon System's 2.4GHz Class F Power Amplifier (PA) implemented in TSMC 180nm. The PA is designed to operate at 2.4GHz, prioritizing high Power Added Efficiency (PAE) for a low power On-Off-Keying (OOK) Transceiver. By shaping the drain voltage and current waveforms via harmonic termination, the Class F topology minimizes the overlap between voltage and current, theoretically pushing efficiency toward 100%. This report covers the full design cycle from component sizing and harmonic matching to post-layout simulations and the final test plan.

2 Key Specifications & Design Requirements

The target specifications for the 2.4GHz Class F PA are summarized below:

- **Technology Node:** TSMC 180nm
- **Supply Voltage (V_{DD}):** 1.2V
- **Operating Frequency:** 2.4 GHz
- **Low Power Output (P_{out}):** $<$ -0.23 dBm
- **High Power Output (P_{out}):** $>$ 10 dBm
- **Power Added Efficiency (PAE):** $>$ 45%
- **Harmonic Termination:** Short circuit at even harmonics, open circuit at odd harmonics.

3 Architecture & High-Level Description

The design consists of three primary stages: the input matching network (IMN), the active transistor core (driver and power stages), and the output matching and harmonic tuning network (OMN).

The IMN is designed to match the 50 Ω source to the input impedance of the driver stage, ensuring maximum power transfer. The IMN also features a Balun off-chip, which converts the single ended PLL output into a differential signal which is sent on chip. All of the circuitry on-chip is pseudo differential. The drivers are CMOS inverters driven rail to rail to buffer the signal and maintain voltage swing before the high impedance Class-F stage. The Class-F stage is implemented as a cascoded common-source, but it is driven well into triode and subthreshold. The cascode device serves primarily to allow for a higher VDD usage to push higher efficiency without leading to device breakdown. Finally, the OMN serves a dual purpose: it matches the optimal load impedance (Z_{opt}) determined via load-pull simulations to the $50\,\Omega$ antenna, and it provides the necessary multi-harmonic terminations (shorting the even harmonics and opening the 3^{rd} harmonic) to approximate square-wave voltage and half-wave rectified current at the drain.

4 Design Process

4.1 Ratiometric Sizing & Biasing

- **Power Stage (M_{power}):** Sized at $W/L = [140.8\mu m/180nm]$, with [32] fingers to minimize gate resistance and maximize f_{max}.
- **Driver Stage (M_{driver}):** Sized ratiometrically at approximately 1/10th the size of the power stage ($W/L = [15\mu m/180nm]$) to provide sufficient buffering while preventing premature compression.
- **Biasing:** The gate bias was set to $V_{GS} = 0.7V$, placing the device near threshold to operate in a Class B-like conduction angle, optimizing the generation of the necessary harmonic content.

4.2 Harmonic Matching Network Design

The core of the Class F operation relies on the output network. Resonant LC tanks at the first and third harmonic implemented to control the fundamental and harmonic impedances.

- 3^{rd} **Harmonic (7.2 GHz):** A parallel LC network provides a high impedance (open circuit) at 7.2 GHz, squaring the voltage waveform.
- 1^{st} **Harmonic (2.4 GHz):** An LC tank resonant at 2.4 GHz was used as a shunt to ground to provide a short circuit for all frequencies other than the fundamental which we wish to output.

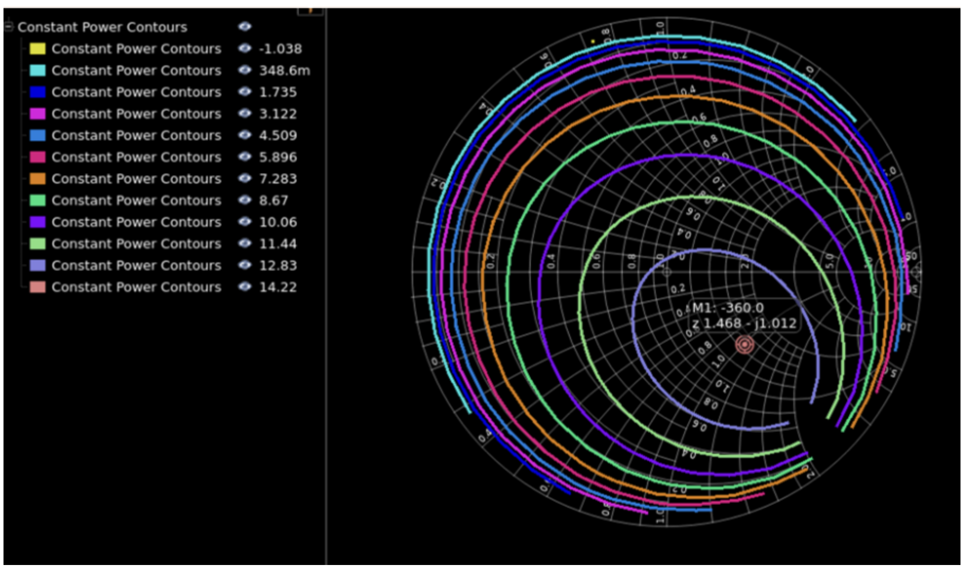
5 Simulation Results

5.1 DC & Stability Analysis

DC simulations were run to verify the bias points across process, voltage, and temperature (PVT) corners. Small-signal S-parameter simulations confirm unconditional stability ($K > 1$ and $B_1 > 0$) from 0-20 GHz.

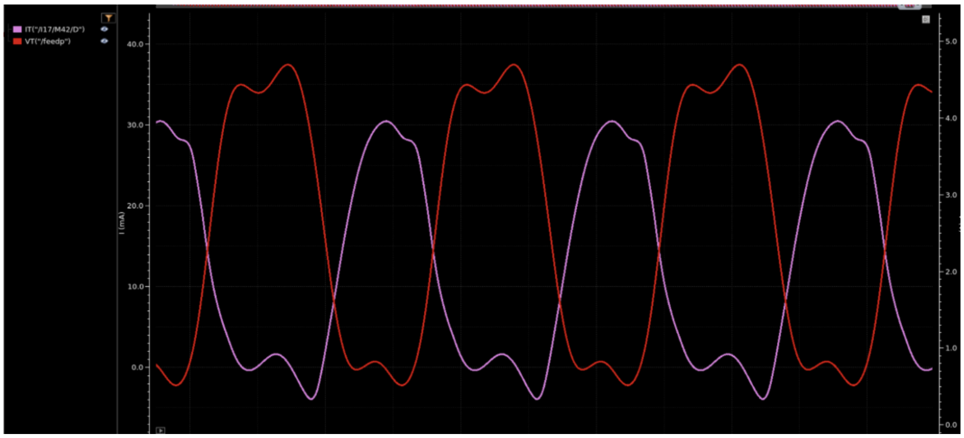
5.2 Load-Pull Analysis

Load-pull simulations were performed at the fundamental frequency to determine Z_{opt} for maximum PAE and maximum P_{out}. This load-pull takes into account the effects of all the transistor parasitics and the harmonic tuning network. The output matching network matches this Z_{opt} to the 50Ω antenna to achieve maximum PAE.

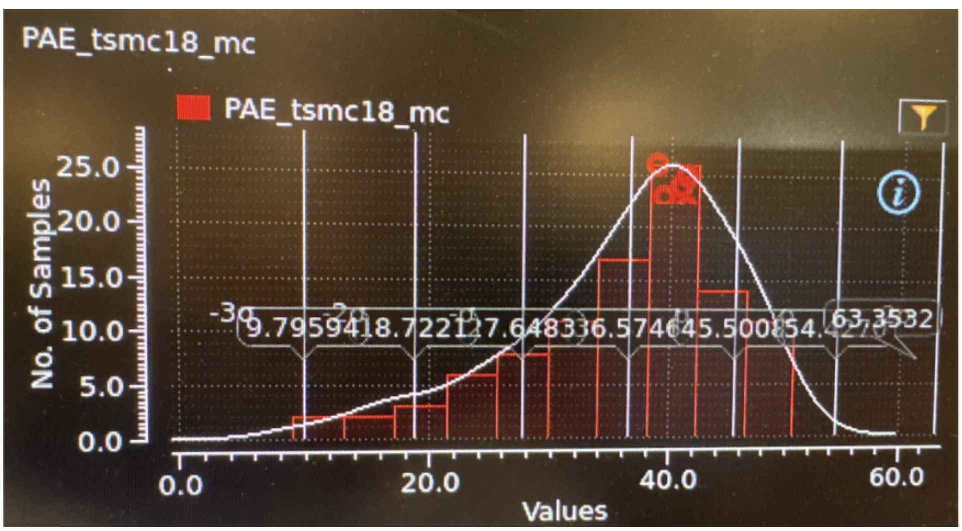


5.3 Transient Waveforms

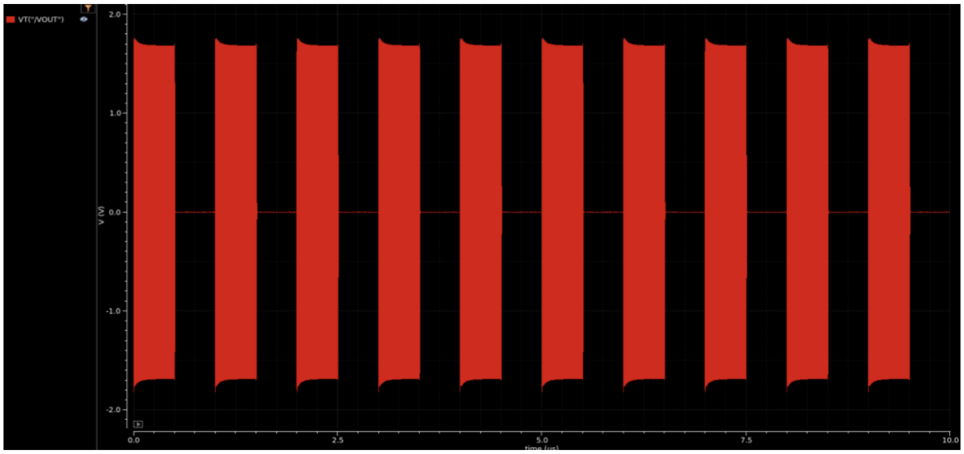
To confirm Class F operation, the drain voltage and current waveforms were plotted over two RF cycles. The voltage waveform demonstrates the expected ”squaring” due to the 3rd harmonic, while the current waveform approximates a half-rectified sine wave.



5.4 Monte-Carlo & PVT Corners



5.5 On-Off Keying Modulation



6 Pinout and Platforms Information

This section should include a table of all the pins, their numbers, purpose, and what to connect them to.

It should also have a diagram and detailed description of all of the off chip infrastructure needed to ensure proper functionality and testability of the power amplifier.

7 Testing & Characterization Plan

Upon receiving the chip, testing will be completed on the 2026 RF board designed by the Platforms subteam. It is imperative that proper communication between RFIC and Platforms be maintained when designing the board to prevent errors that make testing difficult. The below list is a step by step guide for testing the Power Amplifier. Some specifics are omitted as they will be PCB dependent. This document should be updated to reflect those changes.

1. **DC Verification:** Safe power-up sequence, verifying current draw on the V_{DD} and bias lines.

2. **Single-Ended Testing:** The PLL signal should be sent into the

3. **Differential Testing:**